SE444: Introduction to Artificial Intelligence
Final Exam – Grading Key with Minimum Solutions & Rubrics

Monday, December 22, 2025 — Total: 30 points — Duration: 120 minutes

Part I: Multiple Choice Questions (6 pts, 30 min)

Question	1	2	3	4	5	6
Answer	D	B, E	B	E	B	B
Points	1	1	1	1	1	1

Rubric: 1 point each question.
Q2 (multi-select): +0.5 per correct answer, −0.25 per wrong selection. Minimum score: 0.

Quick Reference:

- **Q1 (d-Separation):** D – $B \perp E$ because two unobserved colliders (C and D) block the path between B and E.
- **Q2 (Bayes):** B & E – Calculate $P(G|M) \approx 0.142$ which is between 0.10–0.20, and $P(G|M) < P(M|G)$.
- **Q3 (MDP):** B – Under Action-A only policy, $V(s) = 10$. Note: optimal is B with $Q(B) = 14$.
- **Q4 (Game Theory):** E – Column player has dominant strategy Right (2¿1 and 3¿0).
- **Q5 (STRIPS):** B – Visited(WP1) is added; At(Base) and Cleared(WP1) are deleted.
- **Q6 (Code):** B – Knowledge-based agent pattern (uses kb.tell/ask).

Part II: Robot Navigation Under Uncertainty (16 pts, 60 min)

Q II.1: BN Robot Perception (4 pts, 15 min)

(a) Draw BN Structure [2 pts]

Minimum Answer:
Nodes: O (Obstacle), C (Camera), L (Lidar), W (Weather), P (PathBlocked), A (Alert)

Edges:

- $O \to C$ (Obstacle causes Camera detection)
- $O \to L$ (Obstacle causes Lidar detection)
- $O \to P$ (Obstacle causes PathBlocked)
- $W \to C$ (Weather affects Camera accuracy)
- $W \to P$ (Weather causes PathBlocked)
- $P \to A$ (PathBlocked triggers Alert)

Colliders: C (has parents O, W) and P (has parents O, W)

Rubric: 1 pt for correct 6-node structure with edges, 1 pt for identifying both colliders (C and P).

(b) Is $L \perp A \mid P$? [1 pt]

Minimum Answer:
YES, L and A are conditionally independent given P.
Reasoning: The only path from L to A is:
$L \leftarrow O \to P \to A$
When P is observed, this chain is **blocked** at P.

Rubric: 1 pt for correct answer with path tracing explanation.

(c) Colliders & Explaining Away [1 pt]

Minimum Answer:
Colliders: C and P
Explaining Away Example:
Before observing P, O and W are independent.
After observing $P = True$: If we learn $W = True$ (bad weather), this **explains** the blocked path, making obstacle **less likely**.
$P(O = T \mid P = T, W = T) < P(O = T \mid P = T)$

Rubric: 0.5 pt for identifying both colliders, 0.5 pt for explaining away example.

Q II.2: Multi-Sensor Fusion (4 pts, 15 min)

(a) After Camera Detection [2 pts]

Minimum Answer:
Step 1: Compute P(Camera=Detect)

$$P(D) = 0.8 \times 0.2 + 0.15 \times 0.8 = 0.16 + 0.12 = 0.28$$

Step 2: Apply Bayes' Rule

$$P(O|D) = \frac{P(D|O) \cdot P(O)}{P(D)} = \frac{0.8 \times 0.2}{0.28} = \frac{0.16}{0.28} = \boxed{0.571}$$

Rubric: 1 pt for correct Bayes' setup with total probability, 1 pt for correct calculation.

(b) After Lidar Detection [2 pts]

Minimum Answer:
New Prior (from part a): $P(O) = 4/7 \approx 0.571$, $P(\neg O) = 3/7$
Step 1: Compute P(Lidar=Detect) with new prior

$$P(L) = 0.9 \times \frac{4}{7} + 0.1 \times \frac{3}{7} = \frac{3.6 + 0.3}{7} = \frac{3.9}{7}$$

Step 2: Apply Bayes' Rule

$$P(O|Both) = \frac{0.9 \times 4/7}{3.9/7} = \frac{3.6}{3.9} = \frac{12}{13} = \boxed{0.923}$$

Rubric: 1 pt for using part (a) result as new prior, 1 pt for correct final calculation.

Q II.3: Sensor-Based MEU Decision (5 pts, 15 min)

(a) Compute $P(Obstacle|Alert)$ [2 pts]

Minimum Answer:
Step 1: Total Probability

$$P(Alert) = 0.85 \times 0.3 + 0.10 \times 0.7 = 0.255 + 0.07 = 0.325$$

Step 2: Bayes' Rule

$$P(O|Alert) = \frac{0.85 \times 0.3}{0.325} = \frac{0.255}{0.325} = \boxed{0.785}$$

Rubric: 1 pt for correct setup with total probability, 1 pt for correct calculation.

(b) Expected Utility & MEU Action [3 pts]

Minimum Answer:
Using posterior: $P(O|Alert) = 0.785$, $P(\neg O|Alert) = 0.215$
EU(Cross):

$$EU = 0.785 \times (-50) + 0.215 \times 80 = -39.25 + 17.2 = \boxed{-22.05}$$

EU(Wait):

$$EU = 0.785 \times 10 + 0.215 \times 10 = 7.85 + 2.15 = \boxed{+10}$$

MEU Decision: **Wait** because $10 > -22.05$
Reasoning: High collision risk (78.5%) makes crossing too dangerous.

Rubric: 1 pt for each EU calculation (2 pts total), 1 pt for correct MEU decision with reasoning.

Q II.4: Project Code Trace – Bayesian Reasoning (3 pts, 15 min)

(a) Trace compute_evidence() [1 pt]

Minimum Answer:
`compute_evidence(prior=0.4, likelihood_h=0.9, likelihood_n`

$$evidence = (0.9 \times 0.4) + (0.15 \times 0.6) = 0.36 + 0.09 = \boxed{0.45}$$

(b) Trace bayes_update() [1 pt]

Minimum Answer:
`bayes_update(prior=0.4, likelihood=0.9, evidence=0.45)`

$$posterior = \frac{0.9 \times 0.4}{0.45} = \frac{0.36}{0.45} = \boxed{0.8}$$

(c) Impact of Wrong Evidence [1 pt]

Minimum Answer:
If `evidence = 0.5` (incorrect):

$$posterior = \frac{0.36}{0.5} = 0.72 \neq 0.8$$

Why it matters: Evidence is the normalizing constant. Wrong evidence leads to incorrect posterior probabilities, causing the robot to make suboptimal decisions.

Rubric: 1 pt each for (a) and (b). Part (c): 0.5 pt for calculation, 0.5 pt for explanation.

Part III: Multi-Robot Cooperation (8 pts, 30 min)

Q III.1: Robot Task Planning – STRIPS (3 pts, 12 min)

(a) Define Pickup Action [1 pt]

Minimum Answer:
`Pickup(robot, victim, loc)`
Preconditions:
- At(robot, loc)
- VictimAt(victim, loc)
- ¬Holding(robot, victim)

Add List: Holding(robot, victim)
Delete List: VictimAt(victim, loc)

Rubric: 0.5 pt for correct preconditions, 0.25 pt for add list, 0.25 pt for delete list.

(b) Forward Search Plan [2 pts]

Minimum Answer:
Initial: At(R1, Base), VictimAt(V1, Zone3), Safe(Zone1)
Plan:
1. `MoveTo(R1, Base, Zone3)` → At(R1, Zone3)
2. `Pickup(R1, V1, Zone3)` → Holding(R1, V1)
3. `MoveTo(R1, Zone3, Zone1)` → At(R1, Zone1)
4. `PutDown(R1, V1, Zone1)` → VictimAt(V1, Zone1) ✓

Goal Achieved: VictimAt(V1, Zone1)

Rubric: 1 pt for correct action sequence, 1 pt for showing state changes.

Q III.2: Robot Coordination Game (3 pts, 10 min)

(a) Identify Dominated Strategies [1 pt]

Minimum Answer:
No strictly dominated strategies exist.
Reason: Each strategy is better in different scenarios:
- R1: Left better if R2 plays Right; Right better if R2 plays Left
- Same logic for R2 (symmetric game)

Rubric: 1 pt for correct analysis showing no dominance.

(b) Find Nash Equilibria [2 pts]

Minimum Answer:
Best Responses (underlined):

R1 \ R2	Left	Right
Left	(0, 0)	(5, 5)
Right	(5, 5)	(0, 0)

Nash Equilibria:
1. **(Left, Right)** – payoff (5, 5)
2. **(Right, Left)** – payoff (5, 5)

Interpretation: Anti-coordination game – robots succeed when choosing opposite lanes.

Rubric: 1 pt for correct best response analysis, 1 pt for identifying both NE.

Q III.3: Project Code – Logic Agent (2 pts, 8 min)

(a) Purpose of tell() vs ask() [1 pt]

Minimum Answer:
kb.tell(sentence): Adds/asserts facts to the knowledge base. Modifies KB.
kb.ask(query): Queries KB to check if something is true. Returns True/False. Does **NOT** modify KB.

Rubric: 0.5 pt for each function explanation.

(b) Trace safe_moves Loop [1 pt]

Minimum Answer:
After first loop: KB has Wall(N), Wall(E), Free(S), Free(W)
After kb.infer(): Rule Free(X) ⇒ Safe(X) fires
⇒ Safe(South), Safe(West) added to KB
Trace:

Direction	ask(Safe(X))	Result
North	False	skip
East	False	skip
South	**True**	append
West	**True**	append

Return: `["South", "West"]`

Rubric: 1 pt for correct trace with final safe_moves list.

Grading Summary

Part I: MCQ	Pts	Score	Part III	Pts	Score
Q1: d-Separation	1		Q III.1: STRIPS Planning	3	
Q2: Bayes' Rule	1		Q III.2: Game Theory	3	
Q3: MDP	1		Q III.3: Logic Agent	2	
Q4: Game Theory	1		**Part III Total**	**8**	
Q5: STRIPS	1				
Q6: Code	1				
Part I Total	**6**				
Part II: Robot Nav	**Pts**	**Score**			
Q II.1: BN Perception	4				
Q II.2: Sensor Fusion	4				
Q II.3: MEU Decision	5		**GRAND TOTAL**	**30**	
Q II.4: Code Trace	3				
Part II Total	**16**				

General Grading Guidelines:

- Award partial credit for correct setup with calculation errors
- Give credit for correct reasoning even if final answer is wrong
- No credit for correct answer without showing work
- Accept equivalent valid mathematical formulations
- For multi-step problems, award proportional credit for correct intermediate steps